

Project Design Phase

Problem – Solution Fit Template

Date	5 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ☐ Solve complex problems in a way that fits the state of your customers.
- ☐ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ☐ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ☐ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ☐ **Understand the existing situation in order to improve it for your target group.**

Template:

Customer	1. CUSTOMER SEGMENT(S) CS - Banks and Financial Institutions - Credit Analysts - Loan Officers - Financial Compliance Officers - Prospective Credit Card Applicants - FinTech Companies Who is your customer? Banks, financial organizations, and customers who want fast and accurate credit card eligibility assessment.	6. CUSTOMER CONSTRAINTS CC - Large number of applications - Manual verification is slow - Human errors during evaluation - Difficulty identifying risky applicants - Need for accurate predictions - Regulatory compliance requirements What prevents preventing your customer from seeking advice or from their choice of solution? Traditional systems require significant human intervention and often lack intelligent predictive capabilities.	5. AVAILABLE SOLUTIONS AS - Manual document verification - Rule-based approval systems - Traditional credit scoring methods - Manual review by credit analysts - Third-party credit assessment services What solutions are available to the customer when they face this problem?	Explore Alternatives
	2. JOBS-TO-BE-DONE / PROBLEMS JtBP - Evaluate credit card applications efficiently - Predict approval or rejection accurately - Reduce manual verification time - Identify high-risk applicants - Improve consistency in decision-making - Reduce financial risk due to poor approvals What pains, headaches or problems is your customer trying to solve? What are the underlying needs?	9. PROBLEM ROOT CAUSE RC - Manual credit evaluation process - Increasing application volume - Complex financial data - Inconsistent approval decisions - Time-consuming verification - Human bias and errors What is the real reason why this problem exists? What is the root cause behind the need to rely on guesswork? Use 5-Why method if required.	7. BEHAVIOUR BE - Analysts manually review applications - Customers wait several days for approval - Banks verify multiple financial documents - Decisions depend heavily on human judgment - High-risk applications require additional review In trying to solve or avoid this problem right now, what is the customer thinking, feeling, saying and doing? What is the current behaviour gap between customer's aspirations and what is happening today?	
Solution	3. TRIGGERS TR - Customer submits a credit card application - High volume of daily applications - Need for faster approval decisions - Manual review delays - Increasing demand for digital banking What triggers the customer to look for a solution? What internal or external events start the journey?	10. YOUR SOLUTION SL Credit Card Approval Prediction System An AI-powered machine learning application that predicts whether a credit card application is likely to be approved or rejected based on applicant financial and demographic information. The system: • Performs automated data preprocessing • Uses feature engineering to improve prediction quality • Trains multiple classification models • Selects the best-performing model • Provides instant approval/rejection predictions through a Flask web application • Supports scalable cloud deployment using IBM Watson Machine Learning How does your product or service solve the customer's problem? What is unique about your solution? What value does it deliver to the customer?	8. CHANNELS OF BEHAVIOUR CH - Flask Web Application - Bank Internal Portal - IBM Cloud Deployment - Financial Institution Dashboard - Web Browser - REST API Integration Through which channels does the customer currently try to solve this problem or get advice?	Measure Outcomes, Not Outputs
	4. EMOTIONS: BEFORE / AFTER EM - Before: Uncertain, stressful, time-consuming, high workload, risk of errors - After: Confident, fast decision, less manual effort, consistent process, satisfied customers What is the customer feeling before and after solving the problem? How does the solution make them feel?	11. METRICS MT - Model Accuracy - Precision, Recall, F1 Score - ROC-AUC Score - Prediction Response Time - Number of Successful Predictions - User Satisfaction - Reduction in Manual Processing Time What metrics will indicate success? How will you measure impact and outcomes?		



Credit Card Approval Prediction Mission: Empowering financial institutions with AI-driven credit card approval prediction to enable faster, more accurate, and data-driven decision-making while reducing manual effort and financial risk.